

Roll No.

TDM-881

B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022

DISASTER MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Disaster. Discuss the impacts of Disasters on Environment.

(CO1, CO3)

(b) Explain the terms Hazards, Risk, Vulnerability and Resilience of Disaster paradigm.

(CO1, CO3)

(c) List the natural disasters. Also discuss the causes and effects of Cyclones.

(CO1, CO3)

P. T. O.

2. (a) What is drought ? Describe the drought prone areas of India. Also suggest suitable plan to mitigate the effects of the droughts. (CO2, CO4)
- (b) Explain the factors behind the frequent occurrences of the Cloud Bursts in the Uttarakhand Hill regions. Also suggest suitable and sustainable solutions to these disasters. (CO2, CO4)
- (c) What is vulnerability ? Discuss the measures for Disaster Mitigation in vulnerable areas of Asia and Africa. (CO2, CO4)
3. (a) Discuss the various epidemic causes by the man-made disasters. (CO1, CO4)
- (b) The Tauktae Cyclone killed 86 people in the worst offshore disaster ever in India. Explain its reasons and also explain how the damage could have been completely avoided. (CO1, CO4)
- (c) Discuss the role of CT in timely forecast and alert against the disasters. (CO1, CO4)
4. (a) Explain an effective plan for rescue and restoration for the community affected by the Disasters. (CO2, CO5)
- (b) Discuss the role of education and training during the emergency stage of the Disaster. (CO2, CO5)
- (c) What are the major reasons for the earthquakes in the Himalayan region ? Also suggest the suitable and feasible practices for mitigating the effects and effective restoration from these disasters. (CO2, CO5)

(3)

5. (a) Explain the role of each National Disaster Management Authority (NDMA), Local Government, Society for the mitigating and restoring from the effects of Disaster. (CO4, CO5)
- (b) Discuss the National Disaster Management Policy. State the challenges to the timely management of disaster. (CO4, CO5)
- (c) What is Disaster Management Act, 2005 ? Discuss the Environmental legislations related to disaster management in India. (CO4, CO5)

TCS-802

Roll No.

B. TECH. (CSE) (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, May, 2022

MOBILE COMPUTING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Intel's marketing team sought to increase marketing's overall business impact through a targeted account-based marketing (ABM) program. Gartner provided expert guidance and peer-based insights to help the client build and expand an innovative ABM program that improved the partnership between sales and marketing and increased marketing's impact on the communications service provider (CSP) segment. Evaluate data management issues for wireless with suitable case study of above mentioned organization. (CO1)

P. T. O.

- (b) Germany's top operator, DT, pursued an "innovation by collaboration strategy". DT partnered with Fon to build a WiFi network to meet increasing connectivity needs. With this network, DT was able to offer differentiated WiFi-based services such as WLAN TO GO and WiFi Calling. Evaluate data replication for mobile computers with suitable case study of above mentioned organization. (CO1)
- (c) COSMOTE wanted to change its public perception and be recognized as a modern and innovative provider. The operator also wanted to avoid huge 3G/4G investments whilst continuing to meet customer connectivity needs. Finally, they were looking to gain market share. Evaluate adaptive clustering for mobile wireless networks with suitable case study of above mentioned organization. (CO1)
2. (a) As Assurity Consulting began the process of shifting its offerings from individual services to full-scale solutions, it knew that finding messaging that would resonate with prospects and clients would be paramount. Gartner helped it craft these messages in a way that would help it successfully pitch its new solutions to clients and prospects. Evaluate CODA file system with suitable case study of above mentioned organization. (CO2)
- (b) Convergent Brazilian operator Oi became the first company in Latin America to offer a ubiquitous WiFi network. Through a savvy adaptation of its WiFi proposition, Oi was able to target all customer types. Most importantly, the operator turned non-subscribers into Oi WiFi users. Evaluate disconnected operations with suitable case study of above mentioned organization. (CO2)

- (c) Yokohama is the largest Chinatown in Japan, hosting over 21 million visitors a year. In order to provide the best visitor experience, Yokohama Trade Union partnered with Fontech. It gave its members a guest WiFi-enabled router that provides free WiFi access to their end customers through a portal. This captive portal is leveraged as a marketing tool to promote Yokohama's offerings and activities. Evaluate mobile agents computing with suitable case study of above mentioned organization. (CO2)
3. (a) TechSee sought to disrupt the customer service industry with a transformative visual engagement technology. Gartner experts and research provided the client with the essential strategic insights they needed to create differentiated messaging, properly position their company in the market, develop a go-to-market strategy and demonstrate the need for computer vision AI. Evaluate security and fault tolerance with suitable case study of above mentioned organization. (CO3)
- (b) SoftBank rapidly deployed a complimentary WiFi network. The operator was faced with the risk of congestion of its 3G network due to the launch of the iPhone 4S. By offloading traffic from its cellular to WiFi in Frastructure, they were able to match the increasing connectivity needs of its customers in a short period of time and cost-effective way. Evaluate transaction processing in mobile computing with suitable case study of above mentioned organization. (CO3)

P. T. O.

- (c) Vodafone targeted the small and medium business segment with a convergent proposition. It was complemented with a professional Guest WiFi service that allowed business owners to offer secure and high-quality WiFi to its customers. Evaluate ad hoc networks with suitable case study of above mentioned organization. (CO3)
4. (a) Unity Communications was selected by AT&T to build, grow, and increase key metrics in business wireless support offshore to provide 24/7 support, geo-redundancy, and create efficiencies for operations. Evaluate localization with suitable case study of above mentioned organization. (CO4)
- (b) In the 10 years since entering the market, Turkcell has firmly positioned itself as a global leader in digital identity with its signature solution. Evaluate MAC issues with suitable case study of above mentioned organization. (CO4)
- (c) Three of the UK's leading operators enabled Mobile Connect to compare customer information, better understand their audience, and help prevent fraud. Evaluate routing protocols with suitable case study of above mentioned organization. (CO4)
5. (a) A private equity company that manages well-known amazon brands need help in handling sales calls, product inquiry, order tracking, and after-sales service. Unity Communications was able to onboard and assume operations. After 90 days, the company reached its daily metrics to respond to customer requests, returns, and note a significant

improvement in customer satisfaction. Evaluate ad hoc on demand distance vector routing (AODV) with suitable case study of above mentioned organization. (CO6)

- (b) An e-commerce company specializing in Arts and Crafts needed help in managing administrative tasks and other functions of the business. Unity Communications assumes operations in just two weeks and provides simple, efficient assistance at an excellent rate. Evaluate optimized link state routing protocol (OLSR) with suitable case study of above mentioned organization. (CO6)
- (c) A well-known IT and Telecommunications company reached out to Unity Communications to help them with a variety of administrative tasks and sales support operations. By establishing a back-office team to address the demands of the project, the company was able to focus its strength on other important business endeavors and staff productivity increased by up to 70% in just 2 months. Evaluate QoS in ad hoc networks applications with suitable case study of above mentioned organization. (CO6)

Roll No.

TCS-821

B. TECH. (BDA, CSE, CC, IS, ML, DS) (EIGHTH SEMESTER) END SEMESTER EXAMINATION, May, 2022

SOFT COMPUTING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

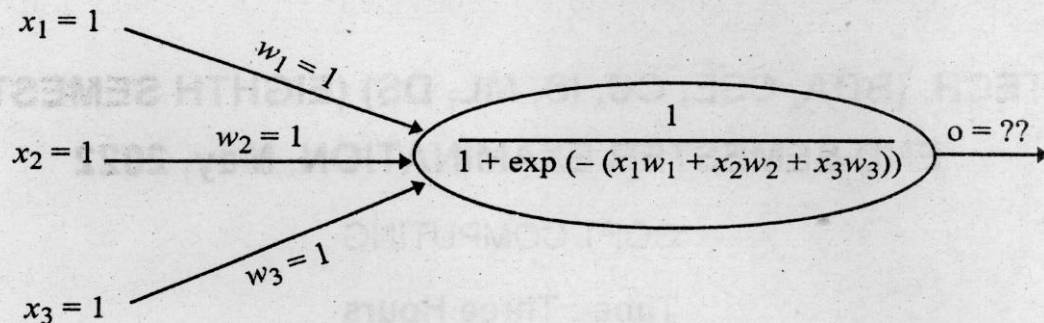
(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Name any *four* different activation functions used in neural network with their pros and cons. (CO2)
- (b) What is perception learning rule ? Mention the limitation of perception algorithm. (CO2)
- (c) Differentiate soft computing from hard computing with the help of suitable examples. (CO1)
2. (a) Consider the single node neural network given below. Let the error function is squared error i. e., $E = |t - 0|$, where $t = 0.5$ is the target output. (CO2, CO3)

P. T. O.

- (i) Calculate the error by the given network. Using Back Propagation weight turning rule, update w_1 , w_2 and w_3 for one iteration only. Assume learning rate $\eta = 0.2$.



- (b) Explain Back propagation algorithm in detail with the help of a diagram. (CO2)
- (c) Propose the architecture of neural network for classifying the samples into three classes named as A, B and C. Assume the dataset contains 1,00,000 images. How many samples you want to use for training and validation ? Give reasons for your answer. (CO2, CO3)
3. (a) Let R and S be the fuzzy relations on the Cartesian space $X \times Y$. Then write the definition for the following operations on the fuzzy set $\mu_R(x, y), \mu_S(x, y)$: (CO4, CO6)
- (i) Union
 - (ii) Intersection
 - (iii) Complement
 - (iv) Containment
- (b) What are the properties of Fuzzy Relations ? Explain in detail. (CO4, CO6)

(3)

- (c) Explain radial basis function network with the help of a diagram. Explain the difference between feed forward neural network and radial basis functions. (CO2)
4. (a) What are the characteristics of Fuzzy Inference Systems ? Explain in detail. (CO4, CO6)
- (b) Explain Mamdani Fuzzy Inference System with the help of a block diagram. (CO4, CO6)
- (c) Compare Mamdani Fuzzy Inference System and Takagi-Sugeno Fuzzy Model. (CO4, CO6)
5. (a) Name and describe the main features of Genetic Algorithms (GA). (CO5)
- (b) Find the values of a, b, c, d which satisfy the equation : (CO5)
- $$a + 2b + 3c + 4d = 30$$
- using genetic algorithm. You can assume the population size is six.
- (c) Find the maximum value of : (CO5)
- $$800 - 62.83(2D + 0.91D - 0.2)$$
- such that $0 < D < 6.3$ using genetic algorithm. You can assume the population size is four. (CO5)

Roll No.

TCS-810

B. TECH. (CSE, CC, IS, ML, DS) (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, May, 2022

VIRTUAL REALITY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Compare Local Illumination and Global Illumination. (CO1)

(b) What is the significance of a haptic system in Virtual Environments ?

Construct a birds eye view of a haptic system using a block diagram.?

(CO1, CO2)

(c) Discuss degrees of freedom in detail. Differentiate between Relative and

Absolute degrees of freedom.

(CO1, CO2)

P. T. O.

2. (a) What are volumetric displays ? Develop a technique to create a hologram. (CO2)
- (b) You want to implement sound reproduction schemes for your virtual reality systems. Evaluate the various options. (CO2, CO3)
- (c) Differentiate between Raster Images and Vector Images. (CO3)
3. (a) What is parallax ? How will you estimate horizontal parallax and what are the different types based on the parallax equation ? (CO3)
- (b) Two virtual reality applications are to be developed : (i) A medical simulation system and (ii) A blind person helper system that provides audio feedback. Give design considerations of these two systems from the viewpoint of isomorphic and non-isomorphic types. Also discuss the pros and cons of the approaches in your design choices for the two systems. (CO3, CO4)
- (c) In modeling virtual environments, explain boundary representations and solid modeling in detail. (CO4)
4. (a) Point out the reasons why an abstract method like Menus are sometimes used in Virtual Reality instead of direct interaction. Summarize the different ways of placing flat menus in VR. (CO4)
- (b) Formulate the issues that comes up with text input in VR systems. (CO5)
- (c) Assess the issues in VR systems that require (i) haptic feedback and (ii) voice commands. (CO5, CO6)
5. (a) What do you understand by Temporal Compliance ? Latency causes a break of temporal compliance, what can be done by a VR programmer to counteract this ? (CO5)

(3)

- (b) You have been asked to develop a VR system to simulate the CS/IT building to a newcomer. Most of the users will be first time users of a Virtual Reality system. Provide the design decisions based on human factors for the development of VR systems. (CO6)
- (c) You are developing a VP system to learn driving a geared motorcycle. Your system will use a HMD. Provide the evaluation and testing mechanism of your system starting from the initial develop phase to the final development phase. (CO6)

Roll No.

TCS-822

B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022
MOBILE APPLICATION DEVELOPMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain Android Architecture briefly. which information do you need before you begin coding a mobile app for a client ? (CO1)
- (b) Which components are necessary for a New Android Project ? Define and explain the Android Framework. (CO1)
- (c) What is Agile Development ? What should I look for in a development team ? (CO1)
2. (a) Provide the important core building blocks of Android. What role does Dalvik play in Android development ? (CO2)

P. T. O.

(2)

- (b) Explain briefly, what is meant by Activities ? Explain Activity Lifecycle briefly. (CO2)
- (c) What is the importance of setting up permission in app development ? Explain Implicit and Explicit Intents. (CO2)
3. (a) What is .apk extension in Android ? What is app sandboxing and what is it for ? (CO3)
- (b) How can you store data in Android and iOS ? Which databases and native solutions are available ? (CO3)
- (c) Explain the AndroidManifest.xml file. Why do you need this ? What is On-Demand Mobile Application ? (CO3)
4. (a) Which scenario can test only on real devices but not on an emulator ? (CO4)
- (b) Name the mobile automation tools that are available in the market. Which is better, manual or automated testing ? (CO6)
- (c) How do you troubleshoot the android application which is crashing frequently ? (CO6)
5. (a) Explain sensors in Android. What is an Augmented Reality (AR) App ? (CO5)
- (b) How do you plan to maintain your app post-launch ? Why is a Product Roadmap important for Mobile App Development ? (CO6)
- (c) Describe the process of publishing apps on Google Play and in Apple's App Store. What do you need to publish an app to each store ? (CO6)

Roll No.

TCS-851

B. TECH. (CSE, CC, IS, ML, DS) (EIGHTH SEMESTER) END SEMESTER EXAMINATION, May, 2022

STORAGE NETWORKS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) (i) What is HBA ? Discuss the benefits of using multiple HBAs on a host.
(ii) Why do formatted disks have less capacity than unformatted disks ? (CO1)
- (b) A large company is considering a storage infrastructure—one that is scalable and provides high availability. More importantly, the company also needs performance for its mission-critical applications. Which storage topology would you recommend (SAN, NAS, IPSAN) and why ? (CO1)
- (c) Describe the core elements in a data center. (CO1)

P. T. O.

2. (a) Describe DAS ? DAS provides an economically viable alternative to other storage networking solutions. Justify this statement. (CO2)
- (b) Explain FO-SAN. Discuss the advantage of FC-SW over FC-AL. (CO2)
- (c) What is virtualization ? Discuss the advantages and disadvantages of in band and out of band implementation. (CO2)
3. (a) Discuss and differentiate RAID 0 and RAID 1. Why is RAID 1 not a substitute for a backup ? (CO3, CO6)
- (b) Explain Network Attached Storage ? How the performance of NAS can be affected if the TCF window size at the sender and receiver are not synchronized ? (CO3, CO6)
- (c) Describe RAID ? What are the benefits of using RAID 3 in a backup application. (CO3, CO6)
4. (a) Name and explain the performance characteristics of the hard disk drive. (CO4, CO6)
- (b) SAN is configured for a backup to disk environment, and the storage configuration has additional capacity available. Can you have a NAS gate-way configuration using this SAN ? Discuss the implications of sharing the backup-to-disk SAN environment with NAS. (CO4, CO6)
- (c) Illustrate different protocols available for transmitting storage data traffic over TCP/IP ? Explain the use of these protocols ? (CO4, CO6)

(3)

5. (a) Define server centric IT architecture and storage centric IT architecture with advantages and limitations. (CO5)
- (b) Explain the importance of Information Lifecycle Management. List the activities in developing the ILM strategy. (CO5)
- (c) Illustrate HOT swap, Hot Spare and RAID controller in implementing RAID. (CO5)

Roll No.

TIT-851(A)

**B. TECH. (IT) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022**

BUSINESS INTELLIGENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Explain any decision support application that you have come across.
And explain its core functions in detail. (CO1, CO2)

(b) Describe the Malcolm Baldrige Performance Excellence Framework.
(CO1, CO2)

(c) Explain Key purpose of using IT in Business. Discuss business process management with consideration of various processes and its explanations. (CO1, CO2)

P. T. O.

2. (a) Can XML data be converted into a structured format ? Explain with an example. (CO1, CO2)
- (b) Discuss two best practices for managing the growth of unstructured data and also discuss solution to storage challenges of unstructured data. (CO1, CO2)
- (c) What can you say about data in an email ? Is it structured, semi-structured, or unstructured ? Give reasons in support of your answer. (CO1, CO2)
3. (a) How is OLTP different from ERR systems and how is ERR different from BI ? (CO3, CO6)
- (b) Explain the role of OLAP tools in the BI architecture. (CO3, CO6)
- (c) Construct an ER Diagram to illustrate the work of a bank. Make all the necessary assumptions. (CO3, CO6)
4. (a) Explain these terms in William H. Inmons definition of a data warehouse : (CO4)
- Subject oriented, integrated, non-volatile and time variant.
- (b) How is a federated database different from a data warehouse ? Explain. (CO4)
- (c) What are the advantages of building a data warehouse ? (CO4)

(3)

5. (a) A company is looking for recruiting graduates for their new business unit. They have asked the HR (Human Resource) team to go ahead with recruitment. What KPI's will be most relevant for measuring the effectiveness of the recruiting process ? (CO1, CO5, CO6)

(b) What is Fact based decision making ? Explain your answer with an example. (CO1, CO5, CO6)

(c) Why is measurement, analysis and knowledge management so important for an enterprise ? Give reasons to support your answer.

(CO1, CO5, CO6)

Roll No.

TCS-859

B. TECH. (CC, IS) (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, May, 2022

SERVICE ORIENTED CLOUD ARCHITECTURE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Capital District Physicians' Health Plan Inc. (CDPHP), a physician-founded and guided health plan, accelerated the processing of unstructured medical data and facilitated the delivery of better service to members using the machine learning capabilities of AWS. CDPHP wanted a more efficient way to extract value from millions of unstructured health records to derive insights and improve care for its members. So CDPHP automated its data processing pipeline using a suite of AWS services, including Amazon Comprehend Medical, to extract key health information. With a more agile data infrastructure on AWS, CDPHP improved its overall efficiency by 60 percent and can

P. T. O.

deliver better care. Evaluate full stack with suitable case study of above mentioned organization. (CO1)

- (b) Dow Jones transformed its data analytics processes using Amazon SageMaker and Amazon S3, saving time, enhancing the customer experience, and driving business value. Evaluate web services framework with suitable case study of above mentioned organization.

(CO1)

- (c) Educational technology company Sumadi delivers near-real-time exam supervision simply and cost effectively after migrating to AWS and was able to scale to support 50,000 concurrent users. Evaluate IoC containers with suitable case study of above mentioned organization.

(CO1)

2. (a) The Orchard vastly improved its speed of innovation using a data synchronization solution powered by Amazon MSK and Amazon OpenSearch Service. Evaluate client side and server side with suitable example of above mentioned organization. (CO2)

- (b) Arm Limited migrated to AWS and modernized its solution for intellectual property design and verification, making it possible for the company to optimize its compute and accelerate speed to market. Evaluate SOA with suitable case study of above mentioned organization. (CO2)

- (c) When the COVID-19 pandemic led to worldwide office shutdowns energy company EDF in the UK used Amazon WorkSpaces to create a secure and reliable work-from-home solution for employees. Evaluate cloud services accessibility with suitable case study of above mentioned organization. (CO2)

3. (a) iFood accelerated innovative initiatives that are improving its customers' experiences by developing an internal training program with the support of AWS Training and Certification. Evaluate SOLID principles with suitable case study of above mentioned organization. (CO3)
- (b) CarbonCure uses AWS to turn data into insights, so that it can help customers reduce embodied carbon from concrete in new construction. Evaluate WCF with suitable case study of above mentioned organization. (CO3)
- (c) Since migrating its digital platform TAO to AWS in 2014, OAT has been able to run hundreds of thousands of tests concurrently and deliver a total of 100 million tests online. Evaluate Cloud service SLA's with suitable case study of above mentioned organization. (CO3)
4. (a) Work management company Smartsheet migrated its on-premises data centers to Amazon RDS and increased reliability, scalability flexibility, and agility, improving performance for customers globally. Evaluate SLA with suitable case study of above mentioned organization. (CO4)
- (b) Volkswagen Financial Services (VWFS) is using Amazon Web Services (AWS) to drive cloud transformation across its organization. In this video, Osvaldo Ramirez, head of the North American region development hub, and Stefan Klünker, global product owner, describe how VWFS uses AWS Training and Certification to strengthen its employees' cloud knowledge and skills. Evaluate the server side application development technology with suitable case study of above mentioned organization. (CO4)

- (c) HBO Max, Warner Media's direct-to-consumer platform offering best-in-class quality entertainment, uses Amazon Web Services (AWS) to scale. Evaluate SOA policy management techniques with suitable case study of above mentioned organization. (CO4)
5. (a) Southeast Iowa Regional Medical Center used AWS Elastic Disaster Recovery to set up a recovery site on AWS for its electronic health record system, improving its recovery time objective by 67 percent. Evaluate web services advantages with suitable case study of above mentioned organization. (CO5)
- (b) Futuready migrated to AWS to gain agility and modernize its architecture, reducing deployment and onboarding time by at least 65 percent. An online insurance broker, Futuready allows customers in Indonesia and Thailand to compare over 500 insurance policies and products. Evaluate SOA policy implementation process with suitable case study of above mentioned organization. (CO5)
- (c) By seamlessly integrating AWS services with third-party tools, Koala cuts development time in half, improves time to market from three weeks to 7 hours, and improves the customer experience. Based in Australia, the company operates an ecommerce platform on which customers can purchase furniture with fast and free delivery. Evaluate namespaces with suitable case study of above mentioned organization. (CO5)

Roll No.

TIT-851

B. TECH. (CSE-DS/ML) & B. TECH. (IT)

(EIGHTH SEMESTER)

END SEMESTER EXAMINATION, May, 2022

BUSINESS INTELLIGENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is your understanding of Business Analytics ? Why do you need core business function ? List few common core business functions.

(CO1, CO2)

(b) What are some of the automated IT Applications that are a great help to your university in its day-to-day functioning ? Explain. (CO1, CO2)

(c) Discuss about the Bespoke IT Application with its advantages and its complexities. (CO1, CO2)

P. T. O.

2. (a) Explain OEM (Object Exchange Model) with the help of an example. (CO1, CO2)
- (b) What according to you is the impact of unstructured data on backup and recovery ? (CO1, CO2)
- (c) Why are Web page said to be unstructured data even though they are defined by HTML language which has a rich structure ? (CO1, CO2)
3. (a) Which System (OLTP /OLAP) has support for concurrency control and recovery and why ? (CO1, CO3)
- (b) State the difference in data model for OLTP and OLAP ? (CO1, CO3)
- (c) How can SI be leveraged to provide a 360-degree perspective on the business ? (CO1, CO3)
4. (a) Can the terms BI and Data warehouse be used interchangeably ? Justify your answer. (CO4, CO6)
- (b) Assume you are a project manager who has been sent to collect business requirements for a retail chain. Give a few examples of business requirement that you would have collected. (CO4, CO6)
- (c) Explain the difference between Entity relationship modeling and dimensional modeling. (CO4, CO6)
5. (a) How is a balanced scorecard different from an enterprise dashboard ? Explain. (CO5, CO6)

(3)

(b) Explain the four components of metric data : Subject, Stratum, Quantum and Application, use an example to explain your answer. (CO5, CO6)

(c) KPI reflect the entire performance of the company and will be cascaded down to the level of individuals. Explain giving an example.

(CO5, CO6)

Roll No.

TPE-008

B. TECH. (ECE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022
SEMICONDUCTOR DEVICES AND MODELING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain conductivity and derive the expression for minimum conductivity in a semiconductor. (CO1)
- (b) Explain the variation of electron concentration with temperature for *n*-type semiconductor. (CO1)
- (c) Explain the following in detail : (CO1)
 - (i) Channel length modulation
 - (ii) Body effect

P. T. O.

(2)

2. (a) A step graded Ge diode, having $N_D = 500 N_A$, acceptor impurities are 2 impurity atoms for every 10^8 atoms at room temperature. Find the barrier potential. Assume $n_i = 2.5 \times 10^{13}/\text{cm}^3$.
Total number of Ge atoms $= 4.4 \times 10^{22}/\text{cm}^3$. (CO2, CO6)
- (b) Explain flatband voltage in detail along with expressions. (CO2, CO6)
- (c) Write short notes on the following : (CO2, CO6)
- (i) Hot electron effect
- (ii) Drain Induced Barrier Lowering (DIBL)
3. (a) Explain SOI structures in detail. Mention the advantages of SOI structures. (CO3)
- (b) Discuss the hot carrier effect and impact ionization in a MOSFET. (CO3)
- (c) Describe the Fin-FET structures and explain its working and advantages in detail. (CO3)
4. (a) Write a SPICE code of resistive load MOSFET circuit for DC analysis. (CO3, CO4)
- (b) Explain the principle of circuit simulation and its objectives. (CO3, CO4)
- (c) Differentiate between partially depleted and fully depleted SOI MOSFETs. (CO3, CO4)
5. (a) Describe the working principle of hetero-junction bipolar transistors. (CO1, CO5)
- (b) Explain HEMTs in detail and differentiate between HBTs and HEMTs. (CO1, CO5)
- (c) Explain three terminal MOS structure with proper band diagram. (CO1, CO5)

Roll No.

TCS-859

B. TECH. (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

SERVICE ORIENTED CLOUD ARCHITECTURE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Unilever was formed in 1930 by the merger of Dutch margarine producer, Margarine Unie and British soap maker, Lever Brothers. Today, the consumer goods giant sells food, home care, refreshments, and personal care products in over 190 countries. Unilever has headquarters in London, United Kingdom and Rotterdam, the Netherlands, and subsidiaries in over 90 countries. The company employs more than 170,000 people. In 2012, Unilever reported more than €51 billion in revenue. Evaluate distributed applications in above mentioned organization. 10 Marks (CO1)

OR

- (b) PhotoVogue is an online photography platform. Launched in 2011 and part of Vogue Italia—which is owned by Condé Nast Italia—it allows upcoming photographers to showcase their work. It also gives them the

P. T. O.

chance to have their photos published in Vogue, or be picked by photography agencies. Each picture that's submitted is carefully reviewed by Vogue Italia's editorial staff, who ensure only the highest quality images appear online. Evaluate web services framework in above mentioned organization. 10 Marks (CO1)

2. (a) Using AWS, Pinterest has tripled its use of storage and compute over just two years without worrying about reliability or scalability. Pinterest provides one of the world's largest visual-bookmarking tools, with more than 200 million users and 2 billion boards. The company has used a variety of AWS services to scale its processing, storage, and data-analysis workloads to help developers focus on delighting customers. Evaluate client side and server side in above mentioned organization.

10 Marks (CO3)

OR

- (b) Automotive technology company Cox Automotive used data from the AWS Well Architected Tool to secure executive sponsorship to invest in increased security across its workloads and to reduce risk. Evaluate cloud services accessibility in above mentioned organization.

10 Marks (CO2)

3. (a) Kasada increased the velocity of deployment and uplifted its DevOps and CI/CD practices by working with AWS Partner CMD Solutions to modernize its applications. Kasada is a bot mitigation company based in Australia. The company is using Amazon EKS to orchestrate Kubernetes workloads, AWS CloudFormation to run infrastructure as

code, and AWS Control Tower to standardize governance. Evaluate SOLID principles in above mentioned organization. 10 Marks (CO1)

OR

- (b) Netflix is the world's leading internet television network, with more than 200 million members in more than 190 countries enjoying 125 million hours of TV shows and movies each day. Netflix uses AWS for nearly all its computing and storage needs, including databases, analytics, recommendation engines, video transcoding, and more—hundreds of functions that in total use more than 100,000 server instances on AWS. Evaluate Cloud service SLA in above mentioned organization. 10 Marks (CO2)

4. (a) Founded in 2008, San Francisco-based Airbnb is a community marketplace with over 7 million accommodations and more than 40,000 unique Experiences for customers to book on the company's website or through its iOS and Android applications. Evaluate the server side application development technology in above mentioned organization.

10 Marks (CO3)

OR

- (b) Comcast uses AWS in a hybrid environment to innovate and deploy features for its flagship video product, XFINITY X1, several times a week instead of once every 12-18 months under its old architecture. Comcast is the world's largest cable company and the leading provider of Internet service in the United States. Evaluate SOA policy management techniques in above mentioned organization.

10 Marks (CO3)

P. T. O.

5. (a) Slack provides a messaging platform that integrates with and unifies a wide range of communications services such as Twitter, Dropbox, Google Docs, Jira, GitHub, MailChimp, Trello, and Stripe. The San Francisco-based company, which launched its eponymous app in February 2014, was started by a small group of Silicon Valley entrepreneurs that include Flickr founder Stewart Butterfield. Privately-held Slack is on Fortune Magazine's "Unicorn List" of startup firms worth \$1 billion or more, with a \$2.8 billion valuation supported by a five percent weekly user growth rate and major brand-name customers including Adobe, Samsung, Intuit, NASA, Dow Jones, eBay, and Expedia. Evaluate IoC containers in above mentioned organization.

10 Marks (CO2)

OR

(b) In anticipation of the rapid growth in FASTag users and transactions on its platform, Paytm Payments Bank Limited decided to leverage AWS in January 2021 for faster scale out. At the end of 2021, the company handled more than 1.8 million average daily FASTag transactions and was the largest issuer of FASTags in India. Evaluate web services advantages in above mentioned organization.

10 Marks (CO3)

Roll No.

TCS-861

B. TECH. (CC, IS, ML, DS) (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, May, 2022

DISTRIBUTED SYSTEMS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) (i) Illustrate the functions of distributed operating systems.

(ii) Discuss the layered and kernel based design approaches of operating systems. (CO1)

(b) Explain the architecture of distributed systems. Discuss the advantages of distributed systems over traditional time sharing systems. (CO1)

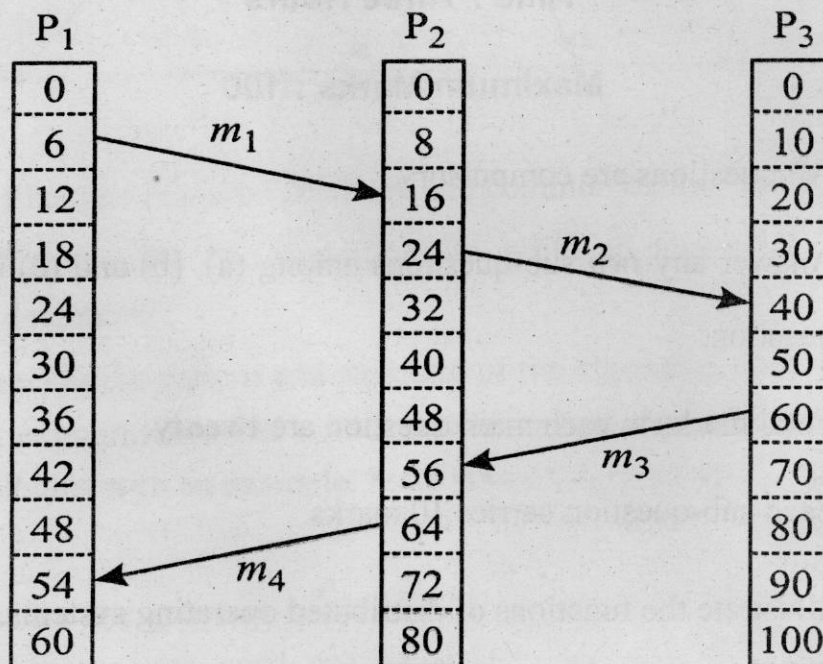
P. T. O.

(c) Name and explain *five* important issues that arise in the design of distributed operating systems. (CO1)

2. (a) Discuss message passing communication model. (CO2)

(b) What are the limitations of distributed systems and their impact on the design and development of distributed systems? (CO2)

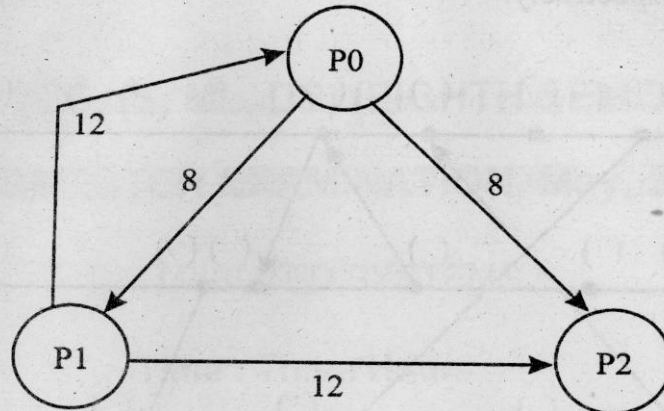
(c) How Lamport synchronizes clocks? Consider a given scenario with three processes. Implement Lamport algorithm to correct the clocks. (CO2)



(Three processes, each with its own clock. The clock run at different rates.)

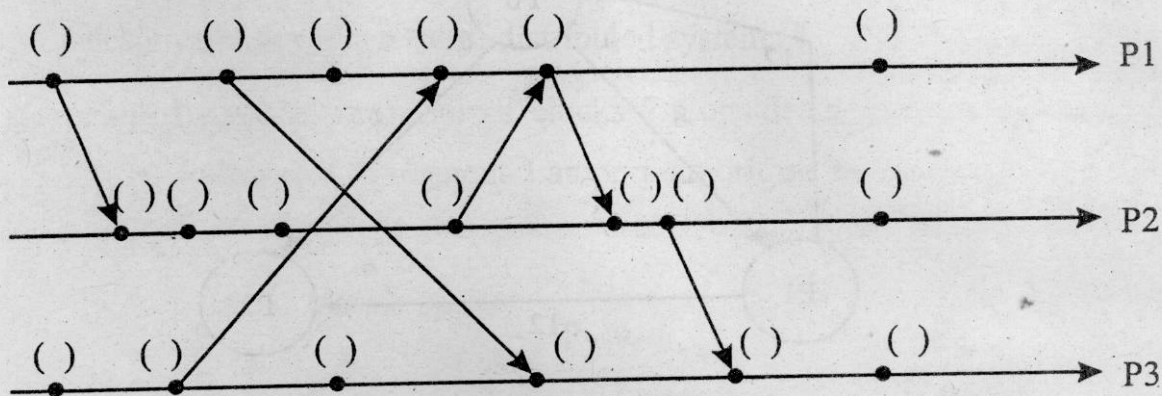
3. (a) Write the basic idea and key steps of the Birman-Schiper-Stephenson causal ordering protocol for causal ordering of messages in distributed systems. (CO3)

- (b) Consider a scenario when process P0 and P1 went to enter critical sector. Both P0 and P1 send broadcast to all 3 processes P0, P1 and P2 in the system along with time stamp. Implement and show all the steps of Ricart – Agarwala’s algorithm for mutual calculation. (CO3)



- (c) “Lamport’s systems of clocks is not powerful enough to capture all the situations.” Justify this statement with an example. (CO3)
4. (a) Give definitions to local state, global state, consistent global state and strongly consistent global state. (CO4, CO6)
- (b) Consider the following algorithm : Each process j maintains a logical clock (that is initially 0) using Lamport’s algorithm. Let T be some integer constant, $T > 0$. When its logical clock advances to or past T for the first time, each process takes a tentative checkpoint. Whenever the process receives a message that was sent at or before T (at the sender) and was received after T (at the receiver), it appends that message in its tentative checkpoint. When the process has received from each neighbor a message that was sent after T , it makes its checkpoint permanent. (Assume that each process communicates with each neighbor infinitely often and that channels are FIFO and reliable.) (CO4, CO6)
- (i) Does the algorithm yield a consistent global checkpoint ? State your assumptions and explain fully.
- (ii) Does the algorithm yield a transitless global checkpoint ? State your assumptions and explain fully.

- (c) For the figure given below, give the values of (i) Lamport's logical clock and (ii) vector clock for each event. The initial values can be assumed as (0) and (0, 0, 0) for Lamport's logical clock and vector clock respectively. (CO4, CO6)



5. (a) (i) Illustrate resource deadlock and communication deadlock.
 (ii) Explain the two conditions to write an algorithm for deadlock detection. (CO5, CO6)
- (b) Following the pattern and structure of the algorithm given for the token ring, explain Raymond's token-based mutual exclusion scheme for tree topologies with an example. State your assumptions and explain fully. (CO5, CO6)
- (c) Illustrate the purpose of a REPLY message in Lamport's algorithm. Note that a site need not necessarily return a REPLY message in response to a REQUEST message. State the condition under which a site does not have to return a REPLY message. Also, give the new message complexity per critical section execution in this case. (CO5, CO6)

(2)

- (b) Compare and contrast ITIL and MoF and name some alternatives of ITIL. (CO1, CO2)
- (c) Discuss about exploratory research to understand customer requirements. (CO1, CO2)
3. (a) Discuss the difference between Service Level Agreement (SLA) and Operational Level Agreement (OLA). (CO3)
- (b) Define the term 'Total Cost of Ownership (TCO) and 'Return on Investment' (ROI). What are significance of these terms in context of IT service management ? (CO3)
- (c) Discuss briefly the *four* important processes involved in the implementation of IT service continuity management. (CO3)
4. (a) What is the difference between incident management and problem management ? (CO4)
- (b) There is a close relationship among change management, configuration management and release management. Justify. (CO4)
- (c) What are the advantages of implementation of change management process in an IT organization ? (CO4)
5. (a) What is the need of storage management ? Explain the process activities of storage management. (CO5, CO6)
- (b) Explain in detail about BMR (Bare Machine Recovery). Give some benefits of BMR. (CO5, CO6)
- (c) Mention various types of privacy laws and discuss information privacy law in detail. (CO5, CO6)

Roll No.

TCS-851

B. TECH. (CSE-CC + IS, DS and ML)

(EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

STORAGE NETWORKS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. A hospital uses an application that stores patient X-ray data in the form of large binary objects in an oracle database. The application is hosted on a UNIX server, and the hospital staff accesses the X-ray records through a Gigabit Ethernet backbone. Storage array provides storage to the UNIX server, which has 6 terabytes of usable capacity. 10 Marks (CO1)

(a) Explain the core elements of a data center.

OR

(b) Describe how the value of this patient data might change over time.

10 Marks (CO1)

P. T. O.

(2)

2. (a) A large company is considering a storage infrastructure—one that is scalable and provides high availability. More importantly, the company also needs performance for its mission-critical applications. Which storage topology would you recommend (SAN, NAS, IPSAN) and why ?
10 Marks (CO2)

OR

- (b) Compare and contrast storage centric architecture and server centric architecture.
10 Marks (CO2)
3. (a) Explain RAID. Illustrate the benefits of using RAID3 in a backup application.
10 Marks (CO3)

OR

- (b) DAS provides an economically viable alternatives to other storage networking solutions. Justify this statement.
10 Marks (CO3)
4. (a) Illustrate the benefits of using RAID 3 in a backup application.
10 Marks (CO3)

OR

- (b) What would you consider while choosing serial or parallel data transfer in a DAS implementation ? Explain your answer and justify your choice.
10 Marks (CO3)
5. (a) Explain the process of data recovery in case of a drive failure in RAID 5.
10 Marks (CO3)

OR

- (b) Why is RAID 0 not an option for Data Protection and high availability ?
10 Marks (CO3)